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Soil Identification and Evaluation

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Purpose

- The purpose of this course is to instruct prospective graduates on the criteria and procedures used to evaluate soil characteristics during Survey and Assault Zone Assessment (SAAZA).



Learning Objectives

- Identify soil classification, type, and characteristics
- Identify the different soil layers
- Assess California Bearing Ratio (CBR) / weight bearing capabilities



References

- MMT TACSOP
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures



References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures



Outline

- Identification
- Distresses
- Evaluation
- Review
- Summary



Outline

- Identification
- Distresses
- Evaluation
- Review
- Summary



Introduction

- Unified Soil Classification System (USCS)
 - System of classification derived from laboratory testing
 - Standardizes the language used to describe soils in survey
 - Used by military and non-military agencies



Introduction

- MMT is not equipped to conduct precision soil identification and evaluation
 - Designed and equipped for contingency operations, hasty evaluation and assessment
- EAF is the primary ACE asset for soil evaluation if precision is required for follow-on operations
 - 3-12 EAF Marines available with MACG Det of a MEU



Soil Identification

- Identification of soil type and characteristics is key to determining the usability of a landing surface
- Base layer only for MMT
- Field identification is our standard

base
subbase
subgrade

ETL 2-19, Appendix A



Soil Identification

- 3 phases to field identification
 - Visual Examination
 - Sedimentation Test
 - Plasticity Tests

ETL 2-19, Appendix A



Soil Identification

- Visual Examination
 - Identify color, grain sizes and approximate gradation, shapes, and properties of undisturbed soil



Soil Identification

- Color



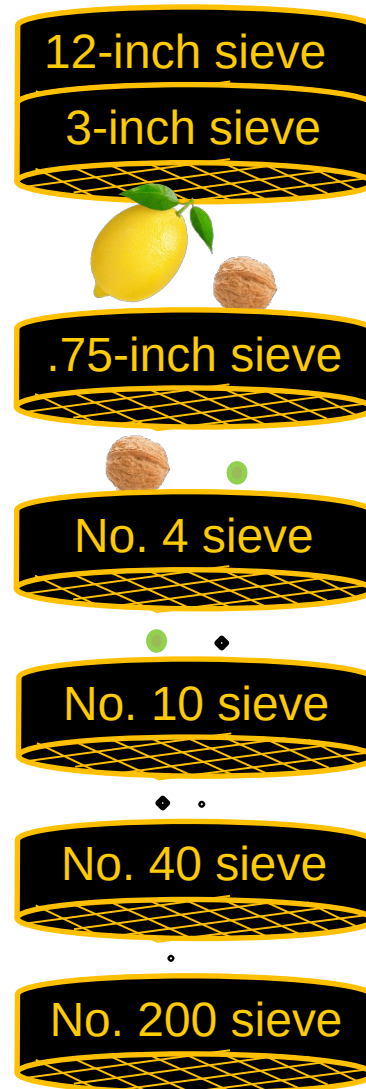


Soil Identification

- Size groups

Visual Examination

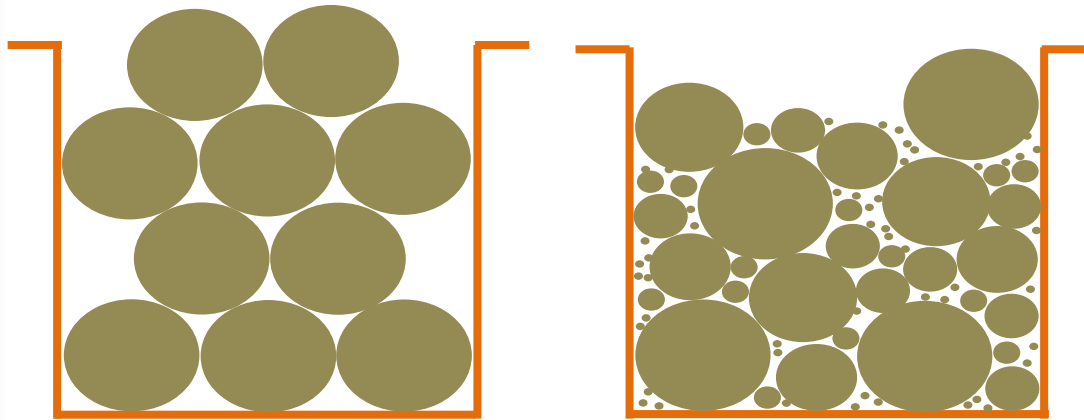
Size Group	Passing	Retained On
Boulders	No max size	12 inch
Cobbles	12 inch	3 inch
Gravels	3 inch	No. 4 sieve
(Coarse)	3 inch	.75 inch
(Fine)	.75 inch	No. 4 sieve
Sands	No. 4 sieve	No. 200 sieve
(Coarse)	No. 4 sieve	No. 10 sieve
(Medium)	No. 10 sieve	No. 40 sieve
(Fine)	No. 40 sieve	No. 200 sieve
Fines	No. 200 sieve	No min size





Soil Identification

- Soil gradation



Poorly-graded

Well-graded





Soil Identification

- Size groups

Visual Examination

Size Group	Passing	Retained On
Boulders	No max size	12 inch
Cobbles	12 inch	3 inch
Gravels	3 inch	No. 4 sieve
(Coarse)	3 inch	.75 inch
(Fine)	.75 inch	No. 4 sieve
Sands	No. 4 sieve	No. 200 sieve
(Coarse)	No. 4 sieve	No. 10 sieve
(Medium)	No. 10 sieve	No. 40 sieve
(Fine)	No. 40 sieve	No. 200 sieve
Fines	No. 200 sieve	No min size





Soil Identification

Major Divisions			Letter Symbol	
Coarse Grained Soils > 50% retained on No. 200 sieve	Gravel and Gravely Soils	< 5% Fines	GW	
			GP	
	>50% retained on No. 4 sieve	> 12% Fines	GM	
			GC	
	Sands and Sandy Soils	< 5% Fines	SW	
			SP	
		< 50% retained on No. 4 sieve	> 12% Fines	SM
				SC
Fine Grained Soils < 50% retained on No. 200 sieve	Silts and Clays LL < 50		ML	
			CL	
			OL	
	Silts and Clays LL > 50		MH	
			CH	
			OH	
Highly Organic Soils			Pt	

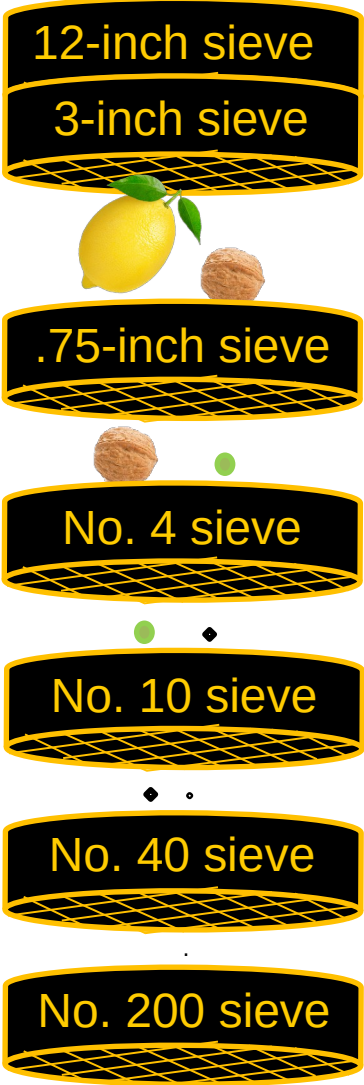
- Three major categories
 - 2 based on grain-size
 - 1 based on presence of organic material
- Subdivided into groups
 - Identified by letter symbols

Groups
– (G) Gravel
– (S) Sand
– (M) Silt
– (C) Clay
Characteristics
– (W) Well-graded
– (P) Poorly-graded
– (H) High compressibility
– (L) Low compressibility
– (Pt) Organic Peat
– (O) Organic Silts and Clays
– Liquid Limits < 50
– Liquid Limits > 50



Soil Identification

Major Divisions			Letter Symbol	
Coarse Grained Soils > 50% retained on No. 200 sieve	Gravel and Gravelly Soils	< 5% Fines	GW	
			GP	
	>50% retained on No. 4 sieve	> 12% Fines	GM	
			GC	
	Sands and Sandy Soils	< 5% Fines	SW	
			SP	
		< 50% retained on No. 4 sieve	> 12% Fines	SM
				SC
Fine Grained Soils < 50% retained on No. 200 sieve	Silts and Clays LL < 50 Silts and Clays LL > 50		ML	
			CL	
			OL	
			MH	
			CH	
			OH	
Highly Organic Soils			Pt	



What do I have the most of by weight in my sample?



Soil Identification

- Determining Soil Type
 - If mostly gravel, runway type is (G)
 - Characteristic is determined by amount and plasticity of Fines in the sample
 - If mostly Sand and Fines, might be (S)
 - Conduct Sedimentation Test to determine if more Sands or more Fines
 - Characteristic determined by amount and plasticity of Fines in the sample
 - If mostly Fines, plasticity test is required to determine characteristic



Soil Identification

- Size groups

Visual Examination

Size Group	Passing	Retained On
Boulders	No max size	12 inch
Cobbles	12 inch	3 inch
Gravels	3 inch	No. 4 sieve
(Coarse)	3 inch	.75 inch
(Fine)	.75 inch	No. 4 sieve
Sands	No. 4 sieve	No. 200 sieve
(Coarse)	No. 4 sieve	No. 10 sieve
(Medium)	No. 10 sieve	No. 40 sieve
(Fine)	No. 40 sieve	No. 200 sieve
Fines	No. 200 sieve	No min size

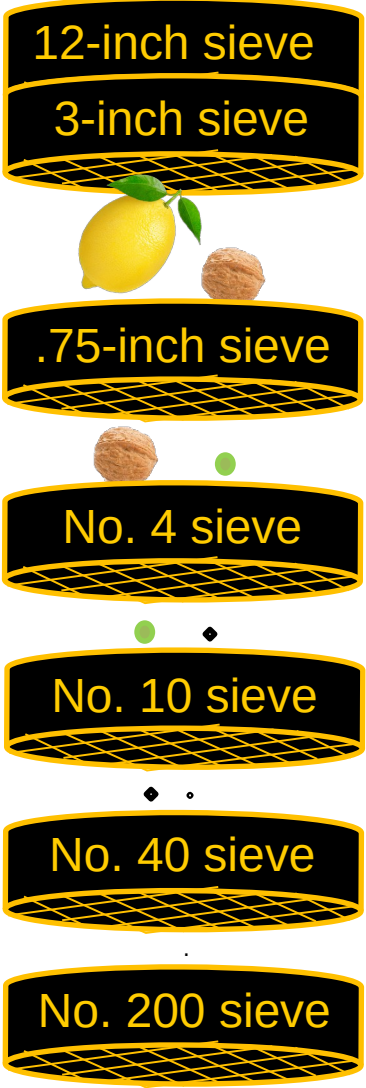
Sedimentation Test





Soil Identification

Major Divisions			Letter Symbol
Coarse Grained Soils > 50% retained on No. 200 sieve	Gravel and Gravely Soils >50% retained on No. 4 sieve	< 5% Fines	GW
			GP
		> 12% Fines	GM
			GC
	Sands and Sandy Soils < 50% retained on No. 4 sieve	< 5% Fines	SW
			SP
		> 12% Fines	SM
			SC
Fine Grained Soils < 50% retained on No. 200 sieve	Silts and Clays LL < 50		ML
			CL
			OL
	Silts and Clays LL > 50		MH
			CH
			OH
Highly Organic Soils			Pt





Soil Identification

- Sedimentation Test



Size Group	Passing	Retained On
Fines	No. 200 sieve	No min size

Settlement time	Grain Diameter	Differentiates
2 seconds	0.4mm	Coarse sand - fine sand
30 seconds	0.072mm (No. 200 sieve)	Sand - fines
10 minutes	0.03mm	Coarse silt - fine silt
1 hour	0.01mm	Silt - clay

Sedimentation Test



Soil Identification

- Determining Percentages
 - Test until 1 hour, or until water is clear and particles are settled
 - All Sands settle within 30 seconds
 - All Fines settle between 30 seconds and 1 hour
 - Characteristic of Fines is determined by Plasticity Test





Soil Identification

- Plasticity Tests

Field Identification of Soils							
Test	Material	Soil Types					
		ML	MH	CL	CH	OL / OH	
Dry Strength	< 40 sieve (wet)	No to Low	Low to Medium	Medium to High	Very high	Low	
Roll / Thread	< 40 sieve (sticky)	Low	Low to Medium	Medium	High	Spongy	
Ribbon	< 40 sieve (sticky)	No cohesion	Little cohesion	3 to 8 inches	8 to 10 inches		
Wet Shake	< 40 sieve (sticky)	Slow to rapid	No to Slow	No to slow	No		
Cast	Damp	Handle freely		Handle roughly			
Bite / Feel	< 40 (< 200) sieve	Unpleasant		Smooth			
Shine		Dull			Shine		
Wash		Discolors quickly, >5% silt					
Dust		> 10% silt					
Sedimentation		30 seconds		1 hour			



Soil Identification

- Plasticity Tests



Puncture Bottle /
Drain Water



Cut Bottle



Allow to Dry
Slightly



Soil Identification

- Dry Strength (Pat) Test
 - Results
 - No strength, crumbles or powders when picked up (ML or MH) or (OL or OH)
 - Low, breaks and powders easily (ML, MH, or CL)
 - Medium, can be broken or powdered with some effort (CL)
 - High, can be broken with great effort, but not powdered (CL)
 - Very High, cannot be broken nor powdered by finger pressure (CH)





Soil Identification

- Dry Strength (Ball) Test
 - Results
 - No strength, crumbles under mere handling (ML)
 - Low, crumbles with some finger pressure (MH)
 - Medium, breaks under considerable finger pressure (MH or CL)
 - High, cannot be broken in hand, but between finger and hard surface (CL or CH)
 - Very High, cannot be broken between thumb and hard surface (CH)





Soil Identification

- Dry Strength (Ball) Test
 - Results
 - No strength, crumbles when touched (GP, SP, SW, GW)
 - Low, withstands careful handling (SM, SC)
 - Medium, can be handled freely (ML or MH)
 - High, withstands rough handling (CL or CH)





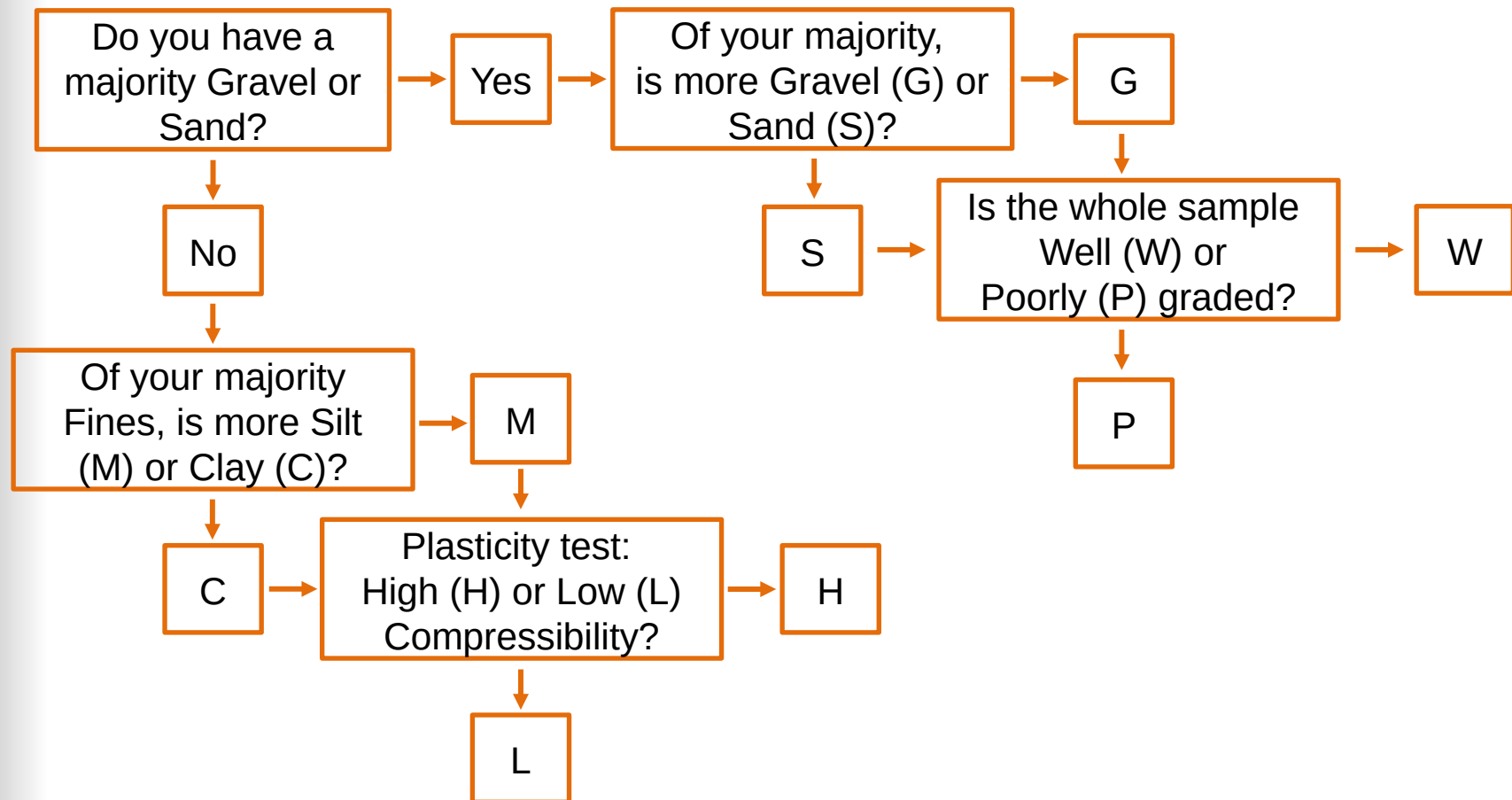
Soil Identification

- Results
 - Majority Gravel, Well-graded (GW)
 - Majority Gravel, Poorly-graded (GP)
 - Majority Sands, Well-graded (SW)
 - Majority Sands, Poorly-graded (SP)
 - Majority Fines, Low Liquid Limit, Low Compressibility (ML, CL, or OL)
 - Majority Fines, Low Liquid Limit, High Compressibility (MH, CH, or OH)



Soil Identification

- Results





Soil Identification

- Groups

- (G) Gravel
- (S) Sand
- (M) Silt
- (C) Clay



Size
Groups

- Characteristics

- (W) Well-graded
- (P) Poorly-graded
- (H) High compressibility
- (L) Low Compressibility
- (Pt) Organic Peat
- (O) Organic Silts and Clays
- (L) Liquid Limits < 50
- (H) Liquid Limits > 50



Size Ratio



Plasticity



Organics
ONLY



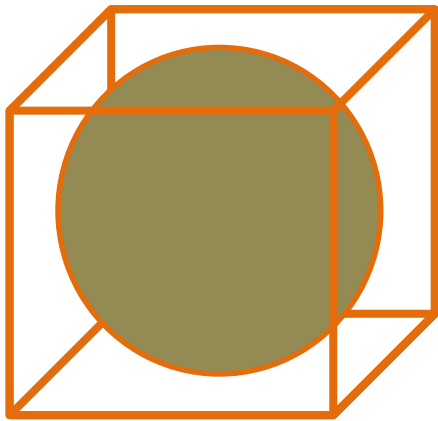
Requires
Lab Tests



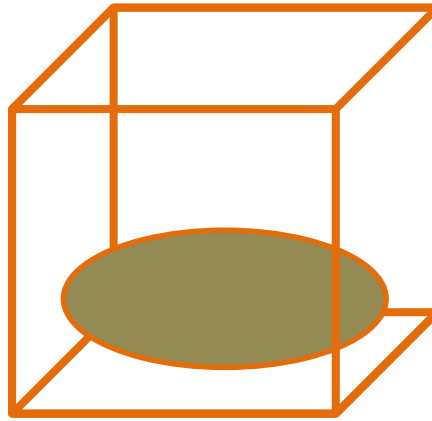


Soil Identification

- Particle Shape



Bulky



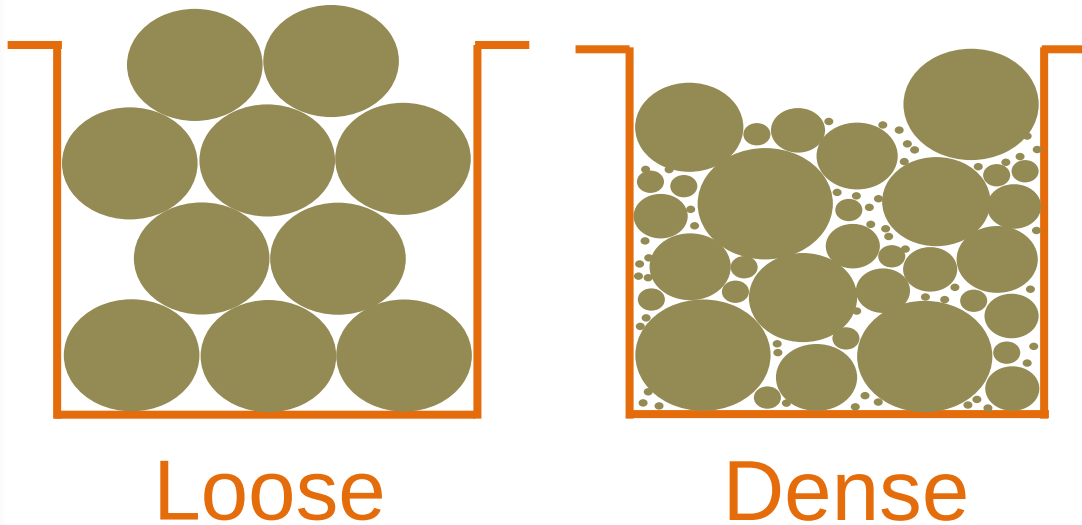
Platy





Soil Identification

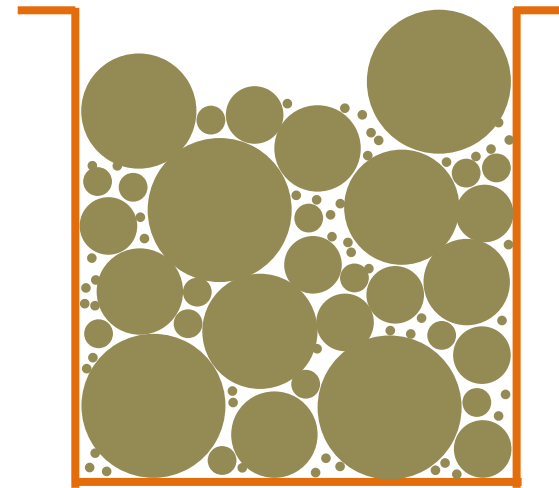
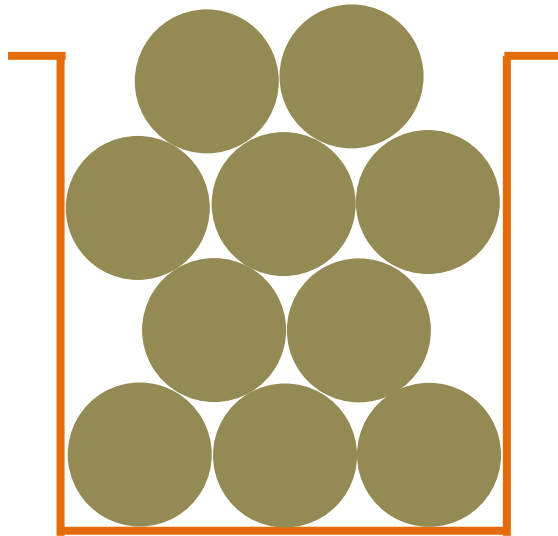
- Compactness





Soil Properties

- Moisture Retention





Soil Properties

- Organic Soils
 - Contain a high content of organic material (plant matter, animal matter)
 - Highly compressible
 - Poor load-maintaining properties
 - Generally unsuitable for landing surfaces, depending on airframe requirements



10 MIN BREAK



Outline

- Identification
- **Distresses**
- Evaluation
- Review
- Summary



Distresses

- Quick and accurate determination of airfield usability is required
 - EAF personnel will use the Semi-prepared Airfield Condition Index (SPACI), which uses a number identifier and assigns a number value

ETL 2-19, Ch 4 Sect 2



Distresses

- Determining airfield distresses

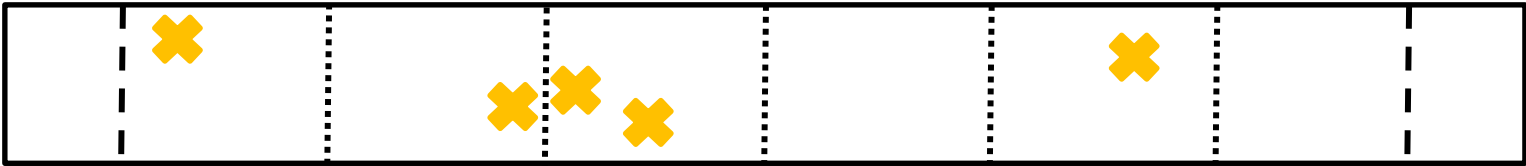
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Distress Severity Levels for KC-130 Semi-prepared Surfaces			
Distress Types	GREEN		RED
Potholes	4" Deep and/or < 15" Diameter	4" to 6" Deep and > 15" Diameter	> 6" Deep and > 15" Diameter
Ruts	Exist but < 4" Deep	4" to 6" Deep	> 6" Deep
Lose Aggregate Coverage	Covers < 1/10 of section	Covers 1/10 to 1/2 of section	Covers > 1/2 of section
Loose Aggregate Size	Max 3/4", Recommended 1/2"	Max 3/4" to 1"	Max > 1"
Dust	Does not obstruct visibility	Partially obstructs visibility	Thick, obstructs visibility
Rolling Resistant Material	Exist but < 1" Deep	1" to 3" Deep	> 3" Deep
Jet Blast Erosion	Exist but < 1" Deep	1" to 3" Deep	> 3" Deep
Stabilized Layer Failure	Exist but < 1" Deep	1" to 3" Deep	> 2" Deep



Distresses

- Potholes (91)

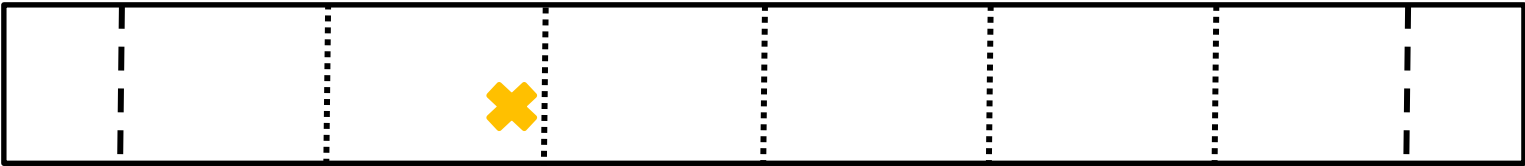


Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Potholes	4" Deep and/or < 15" Diameter		4" to 6" Deep and > 15" Diameter		> 6" Deep and > 15" Diameter



Distresses

- Stabilized Layer Failure (97)

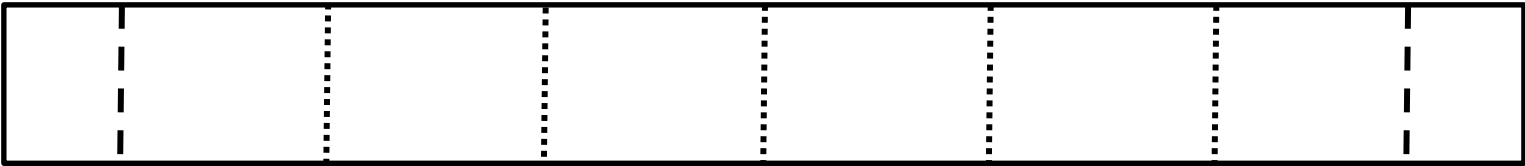


Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Stabilized Layer Failure	Exist but < 1" Deep		1" to 3" Deep		> 2" Deep

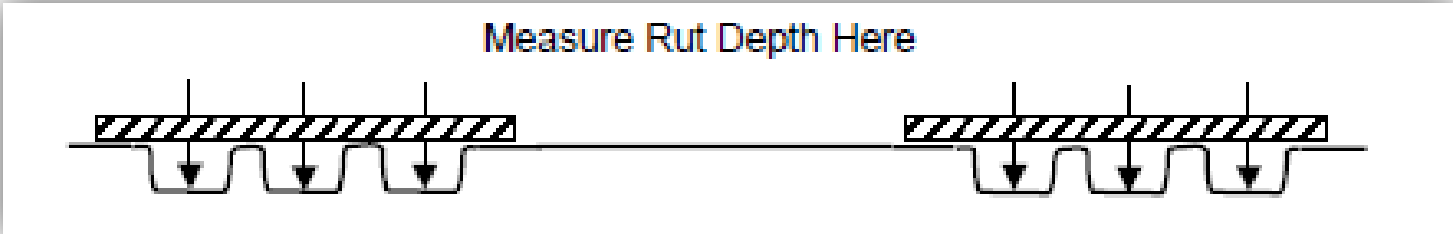


Distresses

- Ruts (92)



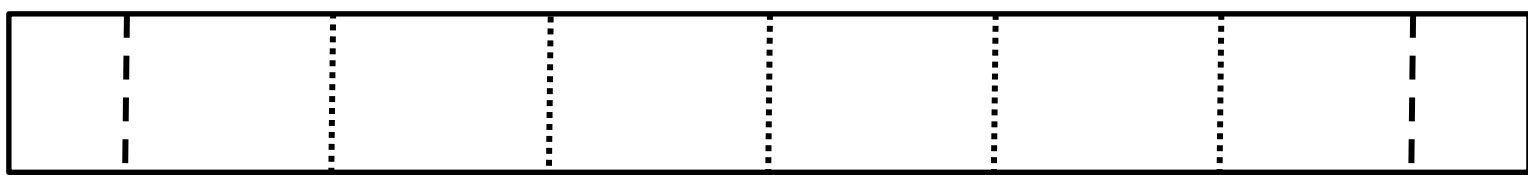
Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Ruts	Exist but < 4" Deep		4" to 6" Deep		> 6" Deep



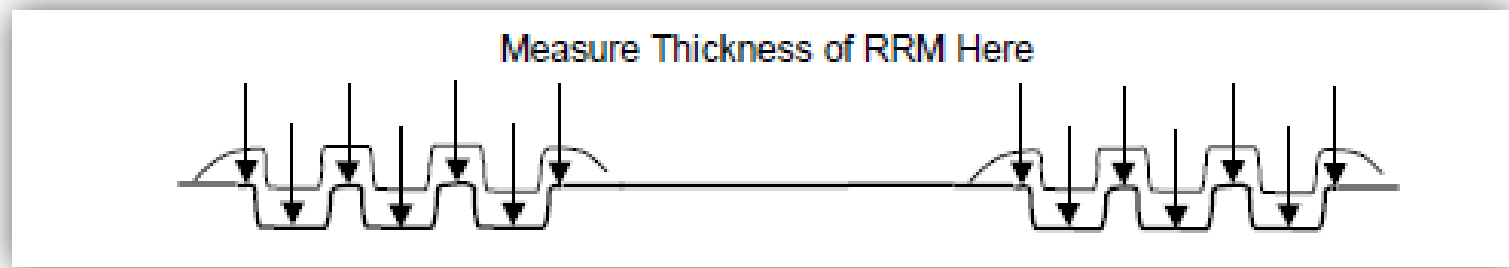


Distresses

- Rolling Resistant Material (RRM) (95)



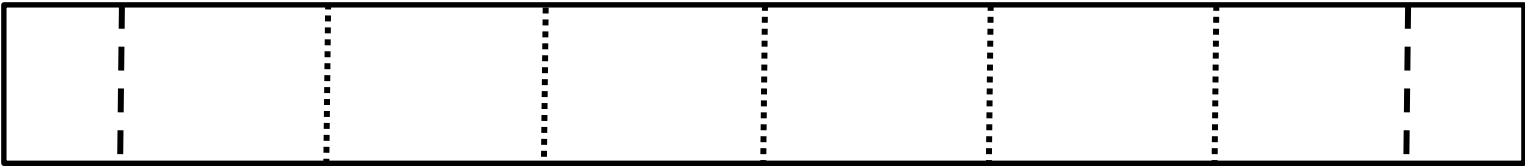
Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Rolling Resistant Material	Exist but < 1" Deep		1" to 3" Deep		> 3" Deep





Distresses

- Loose Aggregate Coverage (93)

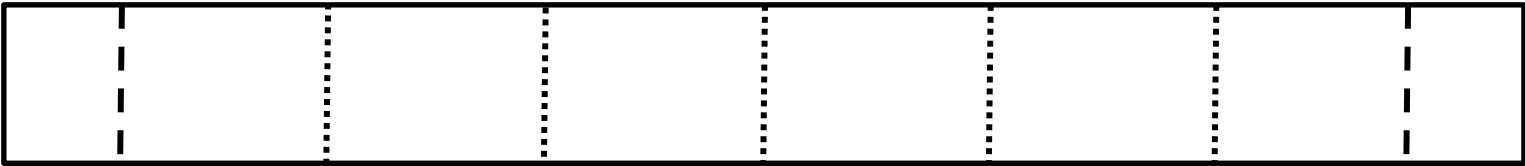


Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Lose Aggregate Coverage	Covers < 1/10 of section		Covers 1/10 to ½ of section		Covers > ½ of section



Distresses

- Loose Aggregate Size (93)

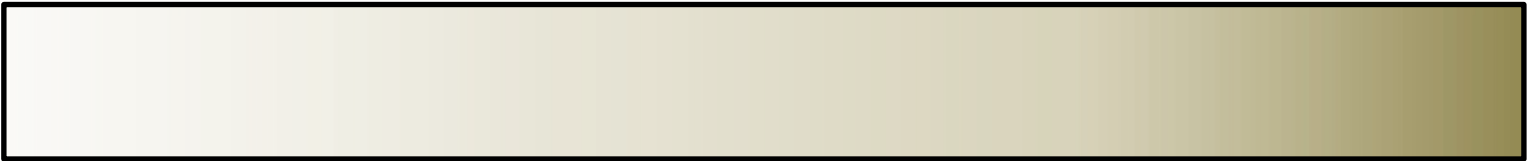


Distress Severity Levels for KC-130 Semi-prepared Surfaces					
	GREEN				
	Max ¾", Recommended ½"				



Distresses

- Dust (94)

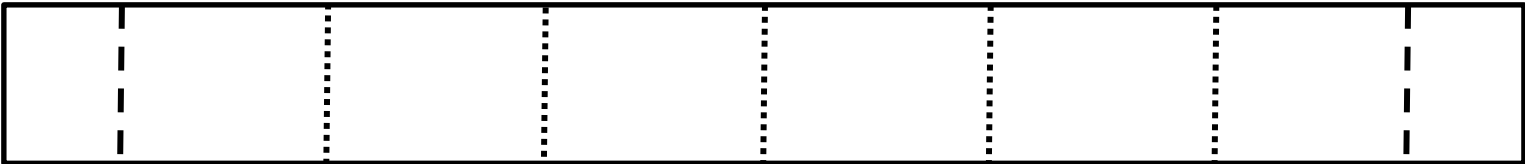


Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Dust	Does not obstruct visibility		Partially obstructs visibility		Thick, obstructs visibility



Distresses

- Jet Blast Erosion (96)



Distress Severity Levels for KC-130 Semi-prepared Surfaces					
Distress Types	GREEN		AMBER		RED
Jet Blast Erosion	Exist but < 1" Deep		1" to 3" Deep		> 3" Deep



Outline

- Identification
- Distresses
- **Evaluation**
- Review
- Summary



Soil Evaluation

Soils		Letter Symbol		Value as Subbase or Subgrade	Value as Base Course	Potential Frost Action	Compressibility and Expansion
Coarse Grained Soils	Gravel and Gravely Soils	GW		Excellent	Good	None to Very Slight	Almost None
		GP		Good to Excellent	Poor to Fair	None to Very Slight	Almost None
		GM	d	Good to Excellent	Fair to Good	Slight to Medium	Very Slight
			u	Good	Poor	Slight to Medium	Slight
		GC		Fair to Good	Poor	Slight to Medium	Slight
	Sands and Sandy Soils	SW		Good	Poor	None to Very Slight	Almost None
		SP		Fair to Good	Poor to Not Suitable	None to Very Slight	Almost None
		SM	d	Good	Poor	Slight to High	Very Slight
			u	Fair to Good	Not Suitable	Slight to High	Slight to Medium
		SC		Fair to Good	Not Suitable	Slight to High	Slight to Medium
Fine Grained Soils	Silts and Clays	ML		Fair to Poor	Not Suitable	Medium to Very High	Slight to Medium
		CL		Fair to Poor	Not Suitable	Medium to High	Medium
	LL < 50	OL		Poor	Not Suitable	Medium to High	Medium to High
	Silts and Clays	MH		Poor	Not Suitable	Medium to Very High	High
		CH		Poor to Very Poor	Not Suitable	Medium	High
	LL > 50	OH		Poor to Very Poor	Not Suitable	Medium	High
Highly Organic Soils		Pt		Not Suitable	Not Suitable	Slight	Very High



California Bearing Ratio (CBR)

- What is CBR?
 - A measure of the soil's weight bearing capacity based on its shearing resistance
 - Shearing resistance is the soil's ability to withstand a given load (shear strength)
 - CBRs are calculated by dividing the pounds of pressure it takes to force a piston (steel rod w/ cone tip) into the earth, by the pounds of pressure it takes to force the same piston to an equal depth into a standard sample of crushed stone in a laboratory



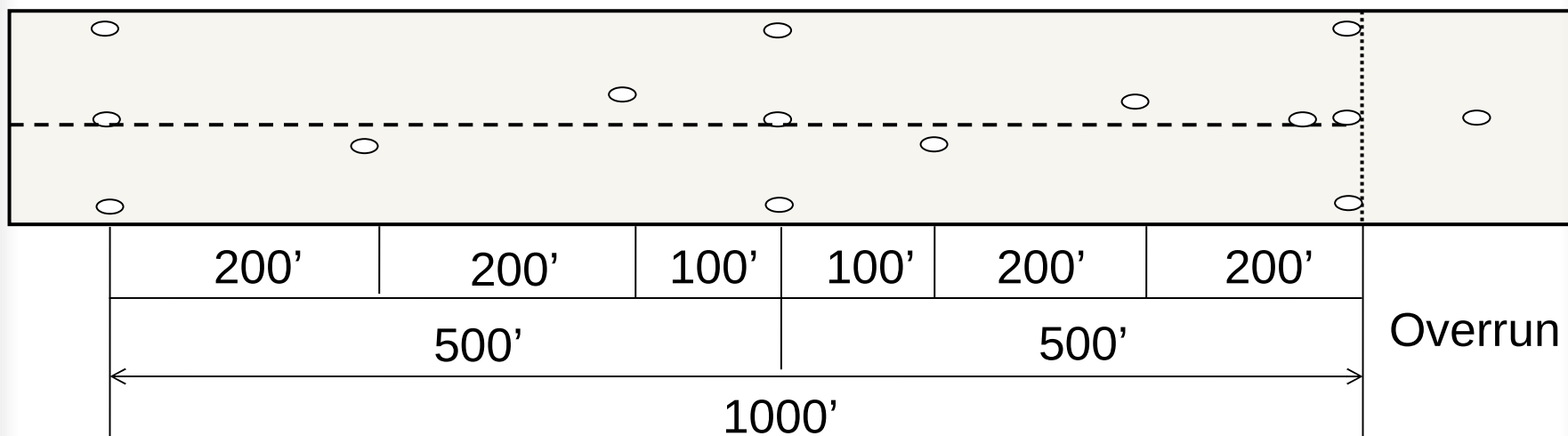
CBR

- The CBR that correlates with a particular type of A/C will determine the number of cycles that the airfield will withstand for that aircraft
- A cycle equals one landing and takeoff



Test Locations

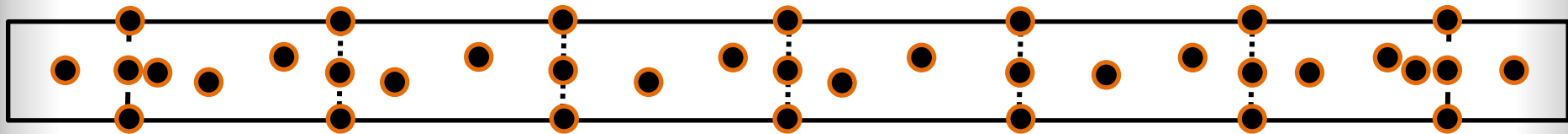
- Overrun (1)
- Approach end CL, L and R (2, 3, 4)
- Bullseye (5)
- 200' R of CL (6)
- 200' L of CL (7)
- 500' CL, L, and R (8, 9, 10)
- 600' R of CL (11)
- 800' L of CL (12)
- 1000' CL, L and R (13, 14, 15)





Test Locations

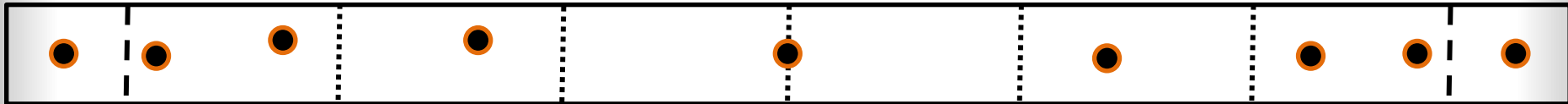
- Repeat pattern for runway length
 - Standard 3000' RWY has 37 locations





Priority Locations

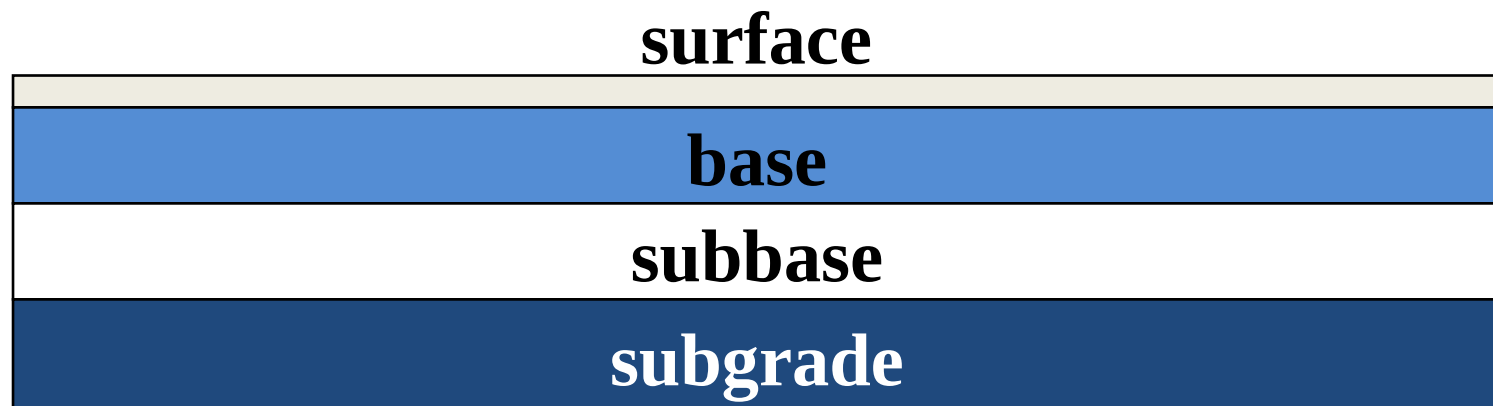
- If time is a factor, focus on critical locations
 - Touchdown / braking / turnaround areas
 - Point of rotation
 - Any identified weak / wet / repaired areas





CBR

- In assault zone survey, we physically measure the soil's shearing resistance with the dynamic cone penetrometer





Dynamic Cone Penetrometer





Purpose and Use

Handle
Hammer

Anvil
Scale
Rod
Cone





Determining CBR

- Dynamic Cone Penetrometer (DCP)
 - Test to a depth of 36"
 - If denial (10 blows w/ o movement) occurs, stop test, relocate 12" and repeat test
 - 3 denials equals CBR 80 for that location
 - If handle deviates more than 5" from plumb, stop test, relocate 6", and repeat test
 - DCP requires 2 personnel to operate
 - Estimate 7 mins per test location for a well-trained team, to include movement
 - Record on DCP Data Sheet for EVERY test



Purpose and Use





Determining CBR

- Dynamic Cone Penetrometer (DCP)
 - Recording Depths
 - Record to nearest .1" (5mm)
 - 1" (25mm) per blow set

ALZ : _____ Date : _____

[illegible]

CBR Layer Structure

Layer Thickness	CBR
150 mm	10
100 mm	15
75 mm	20
50 mm	25
25 mm	30
12.5 mm	35
6.25 mm	40
3.125 mm	45
1.5625 mm	50
0.78125 mm	55
0.390625 mm	60
0.1953125 mm	65
0.09765625 mm	70
0.048828125 mm	75
0.0244140625 mm	80
0.01220703125 mm	85
0.006103515625 mm	90
0.0030517578125 mm	95
0.00152587890625 mm	100



CBR

- Number of hammer blows between test readings
- Accumulative cone penetration after each set of hammer blows
- Difference of (2) at start and end of blow set
- (3) divided by (1)
- From CBR versus DCP correlation

DCP # _____				
Location on ALZ:				
No. of Blows	Accum. Depth	Depth per Blow Set	Depth per Blow	CBR
1	2	3	4	5
0	1.0			
5	2.0	1.0	.2	



CBR

- Record starting point
- Record # of blows
- Record penetration depths
- This information should be recorded during actual test

No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	.5			
3	1.5			
2	2.7			
3	3.8			
3	5.0			
4	6.2			
8	7.5			
10	8.8			
11	10.0			
10	11.3			
11	12.5			
12	13.8			
12	14.9			
14	16.2			
14	17.5			
12	18.8			
10	19.8			
6	21.0			
5	22.2			
5	23.5			
6	24.7			
5	25.9			
4	26.1			
4	27.4			
5	28.6			
5	29.9			
4	31.1			
5	32.4			
4	33.6			

CBR Layer Structure			
Layer	Thickness	CBR	Passes



CBR

- Compute depth per each blow set
 - Difference of (2) at start and end of blow set
- Compute Depth per Blow
 - (3) divided by (1)

No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	.5	0	0	
3	1.5	1	.33	
2	2.7	1.2	.6	
3	3.8	1.1	.35	
3	5.0	1.2	.4	
4	6.2	1.2	.3	
8	7.5	1.3	.1625	
10	8.8	1.3	.13	
11	10.0	1.2	.1091	
10	11.3	1.3	.13	
11	12.5	1.2	.1091	
12	13.8	1.3	.1083	
12	14.9	1.1	.0916	
14	16.2	1.3	.0928	
14	17.5	1.3	.0928	
12	18.8	1.3	.1083	
10	19.8	1.0	.1	
6	21.0	1.2	.2	
5	22.2	1.2	.24	
5	23.5	1.3	.26	
6	24.7	1.2	.2	
5	25.9	1.2	.24	
4	26.1	1.2	.3	
4	27.4	1.3	.325	
5	28.6	1.2	.24	
5	29.9	1.3	.26	
4	31.1	1.2	.3	
5	32.4	1.3	.26	
4	33.6	1.2	.3	

CBR Layer Structure			
Layer	Thickness	CBR	Passes



CBR

- Determine CBR for each Blow Set
 - Use Table 3
 - Tabulated Correlation of DCP Index to CBR on page 19 in ETL 2-19 (next slide)

No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	.5	0	0	N/A
3	1.5	1	.33	25
2	2.7	1.2	.6	14
3	3.8	1.1	.35	25
3	5.0	1.2	.4	22
4	6.2	1.2	.3	30
8	7.5	1.3	.1625	60
10	8.8	1.3	.13	76
11	10.0	1.2	.1091	100
10	11.3	1.3	.13	76
11	12.5	1.2	.1091	100
12	13.8	1.3	.1083	100
12	14.9	1.1	.0916	100
14	16.2	1.3	.0928	100
14	17.5	1.3	.0928	100
12	18.8	1.3	.1083	100
10	19.8	1.0	.1	100
6	21.0	1.2	.2	47
5	22.2	1.2	.24	39
5	23.5	1.3	.26	35
6	24.7	1.2	.2	47
5	25.9	1.2	.24	39
4	26.1	1.2	.3	30
4	27.4	1.3	.325	25
5	28.6	1.2	.24	39
5	29.9	1.3	.26	35
4	31.1	1.2	.3	30
5	32.4	1.3	.26	35
4	33.6	1.2	.3	30

CBR Layer Structure			
Layer	Thickness	CBR	Passes



CBR

DCP Index mm/blow	in/Blow	CBR	DCP Index mm/blow	in/Blow	CBR	DCP Index mm/blow	in/Blow	CBR
<3	0.10	100	12		18	56-57	2.20	3.2
	0.11	92		0.50	17	58		3.1
	0.12	84	13		16	59-60		3.0
3		80	14	0.55	15	61-62	2.40	2.9
	0.13	76	15	0.60	14	63-64	2.50	2.8
	0.14	70	16	0.65	13	65-66	2.60	2.7
	0.15	65	17	0.70	12	67-68		2.6
	0.16	61	18-19		11	69-71	2.80	2.5
4		60	20-21	0.80	10	72-74		2.4
	0.17	57	22-23	0.90	9	75-77	3.00	2.3
	0.18	53	24-26	1.0	8	78-80		2.2
5	0.19	50	27-29		7	81-83		2.1
	0.20	47	30-34	1.20	6	84-87	3.40	2
	0.21	45	35-38	1.40	5	88-89	3.50	1.9
	0.22	43	39		4.8	92-96		1.8
	0.23	41	40		4.7	97-101	4.00	1.7
6		40	41	1.60	4.6	102-107		1.6
	0.24	39	42		4.4	108-114		1.5
	0.25	37	43		4.3	115-121		1.4
7	0.26	35	44		4.2	122-130	5.00	1.3
	0.27	34	45		4.1	131-140		1.2
	0.28	32	46	1.80	4	141-152		1.1
	0.29	31	47		3.9	153-166	6.00	1
8	0.30	30	48		3.8	166-183	7.00	0.9
9	0.35	25	49-50		3.7	184-205	8.00	0.8
	0.40	22	51	2.0	3.6	206-233	9.00	0.7
10-11		20	52		3.5	234-271	10.00	0.6
	0.45	19	53-54		3.4	272-324		0.5
			55		3.3	>324		<0.5

No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	.5	0	0	N/A
3	1.5	1	.33	25
2	2.7	1.2	.6	14
3	3.8	1.1	.35	25
3	5.0	1.2	.4	22
4	6.2	1.2	.3	30
8	7.5	1.3	.1625	60
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10	11.3	1.3	.13	76
11	12.5	1.2	.1091	100
12	13.8	1.3	.1083	100
12	14.9	1.1	.0916	100
14	16.2	1.3	.0928	100
14	17.5	1.3	.0928	100
12	18.8	1.3	.1083	100
10	19.8	1.0	.1	100
6	21.0	1.2	.2	47
5	22.2	1.2	.24	39
5	23.5	1.3	.26	35
6	24.7	1.2	.2	47
5	25.9	1.2	.24	39
4	26.1	1.2	.3	30
4	27.4	1.3	.325	25
5	28.6	1.2	.24	39
5	29.9	1.3	.26	35
4	31.1	1.2	.3	30
5	32.4	1.3	.26	35
4	33.6	1.2	.3	30

CBR Layer Structure			
Layer	Thickness	CBR	Passes



Determining Passes

No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	.5	0	0	N/A
3	1.5	1	.33	25
2	2.7	1.2	.6	14
3	3.8	1.1	.35	25
3	5.0	1.2	.4	22
4	6.2	1.2	.3	30
8	7.5	1.3	.1625	60
10	8.8	1.3	.13	76
11	10.0	1.2	.1091	100
10	11.3	1.3	.13	76
11	12.5	1.2	.1091	100
12	13.8	1.3	.1083	100
12	14.9	1.1	.0916	100
14	16.2	1.3	.0928	100
14	17.5	1.3	.0928	100
12	18.8	1.3	.1083	100
10	19.8	1.0	.1	100
6	21.0	1.2	.2	47
5	22.2	1.2	.24	39
5	23.5	1.3	.26	35
6	24.7	1.2	.2	47
5	25.9	1.2	.24	39
4	26.1	1.2	.3	30
4	27.4	1.3	.325	25
5	28.6	1.2	.24	39
5	29.9	1.3	.26	35
4	31.1	1.2	.3	30
5	32.4	1.3	.26	35
4	33.6	1.2	.3	30

CBR Layer Structure			
Layer	Thickness	CBR	Passes
Surface	.5	N/A	N/A
Base	5.7	22	
Sub-Base	12.3	92	
Sub-grade	12.6	39	

- Determine layers
 - Based on significant change in CBR
- Calculate thicknesses
 - Subtract last depth from previous layer from last depth of current layer
 - Ex: 6.2" minus .5" = 5.7", depth of base layer
- Calculate total blows
 - Add all blows from a respective layer
 - Ex: there are 15 total blows in the base layer we determined



CBR

No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	.5	0	0	N/A
3	1.5	1	.33	25
2	2.7	1.2	.6	14
3	3.8	1.1	.35	25
3	5.0	1.2	.4	22
4	6.2	1.2	.3	30
8	7.5	1.3	.1625	60
10	8.8	1.3	.13	76
11	10.0	1.2	.1091	100
10	11.3	1.3	.13	76
11	12.5	1.2	.1091	100
12	13.8	1.3	.1083	100
12	14.9	1.1	.0916	100
14	16.2	1.3	.0928	100
14	17.5	1.3	.0928	100
12	18.8	1.3	.1083	100
10	19.8	1.0	.1	100
6	21.0	1.2	.2	47
5	22.2	1.2	.24	39
5	23.5	1.3	.26	35
6	24.7	1.2	.2	47
5	25.9	1.2	.24	39
4	26.1	1.2	.3	30
4	27.4	1.3	.325	25
5	28.6	1.2	.24	39
5	29.9	1.3	.26	35
4	31.1	1.2	.3	30
5	32.4	1.3	.26	35
4	33.6	1.2	.3	30

- Determine average CBR for each layer
 - Divide thickness of each layer by total number of blows per layer
 - Base: $5.7'' / 15 = .38''$ per blow
 - Sub-base: $13.6'' / 124 = .11''$ per blow
 - Sub-grade: $13.8'' / 58 = .24''$ per blow
 - Reference ETL 09-6 Table 3

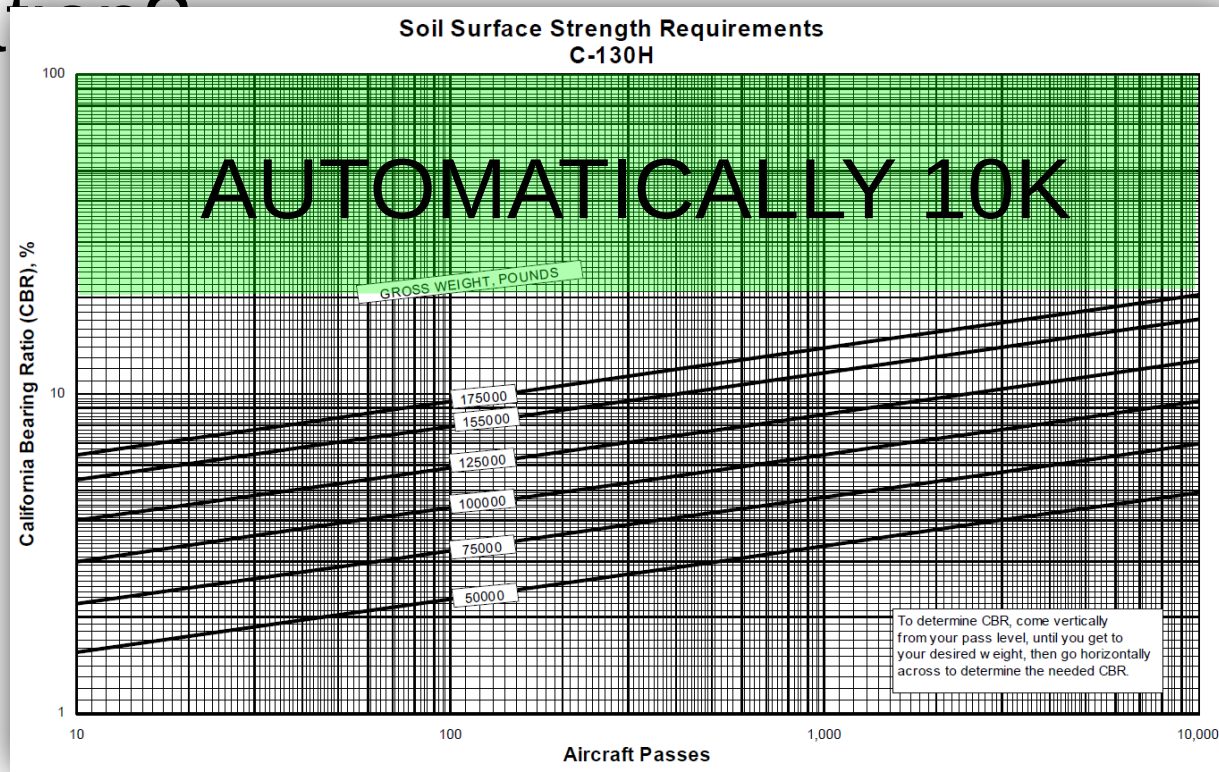
CBR Layer Structure

Layer	Thickness	CBR	Passes
Surface	.5	N/A	N/A
Base	5.7	22	
Sub-Base	13.6	92	
Sub-grade	13.8	39	



Determining Passes

- What is the maximum allowable passes for a 175,000 lbs. KC-130 at the test location?

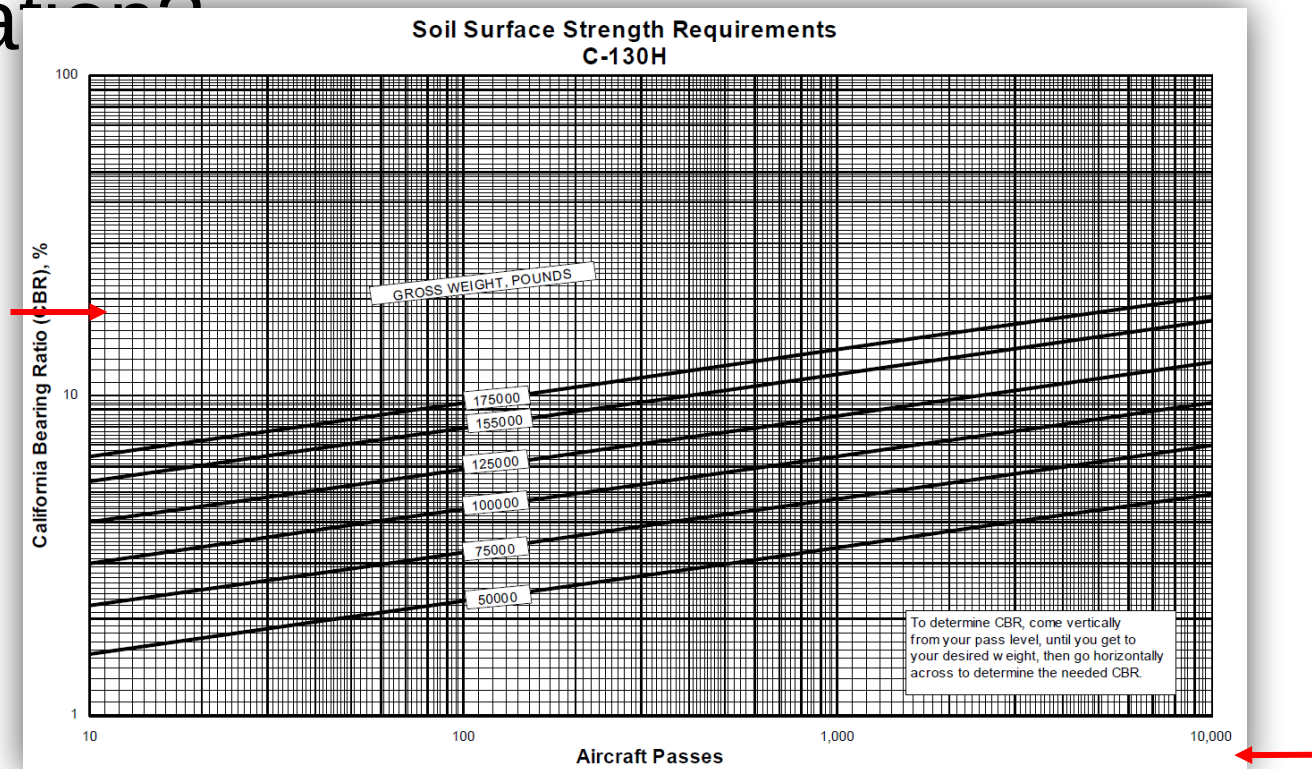


Base: 5.7", CBR 22 – Sub-base: 13.6", CBR 92 – Sub-grade: 13.8", CBR 39



Determining Passes

- What is the maximum allowable passes for a 175,000 lbs. KC-130 at the test location?

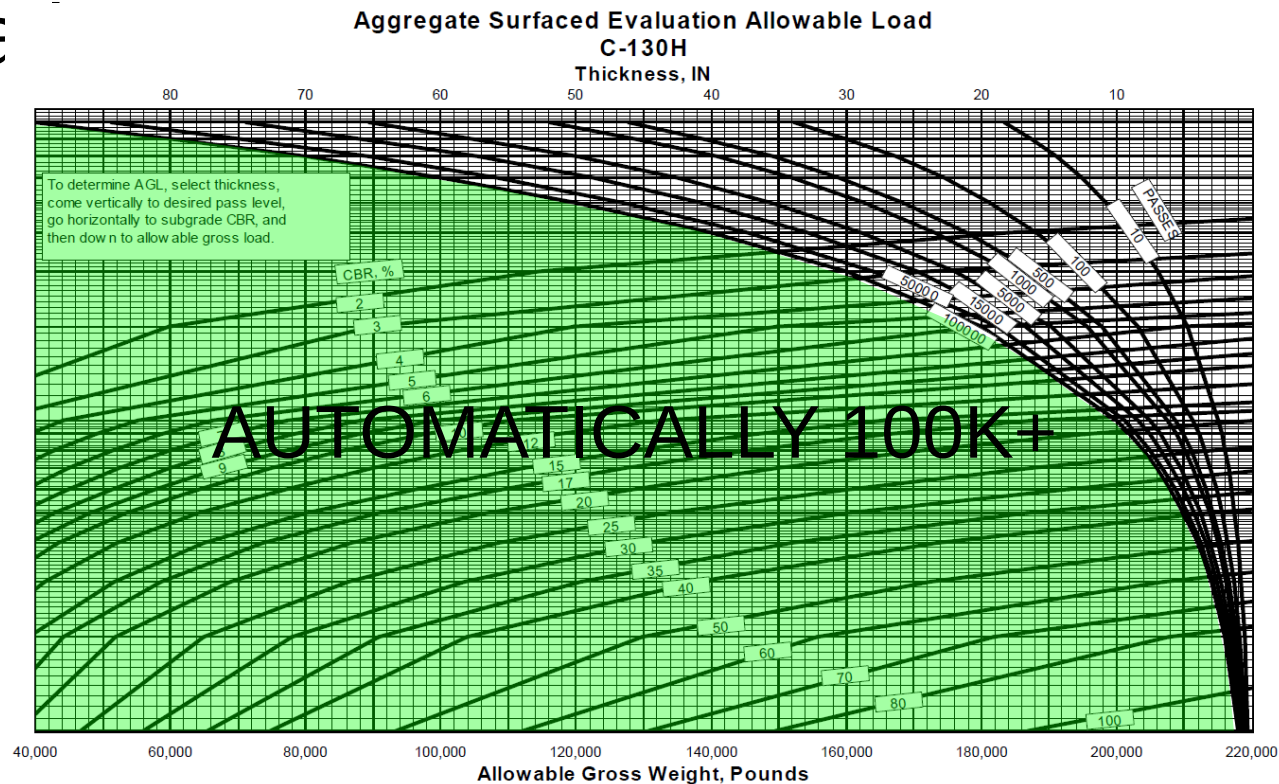


Base: 5.7", CBR 22 – Sub-base: 13.6", CBR 92 – Sub-grade: 13.8", CBR 39



Determining Passes

- What is the maximum allowable passes for a 175,000 lbs. KC-130 at the test loca

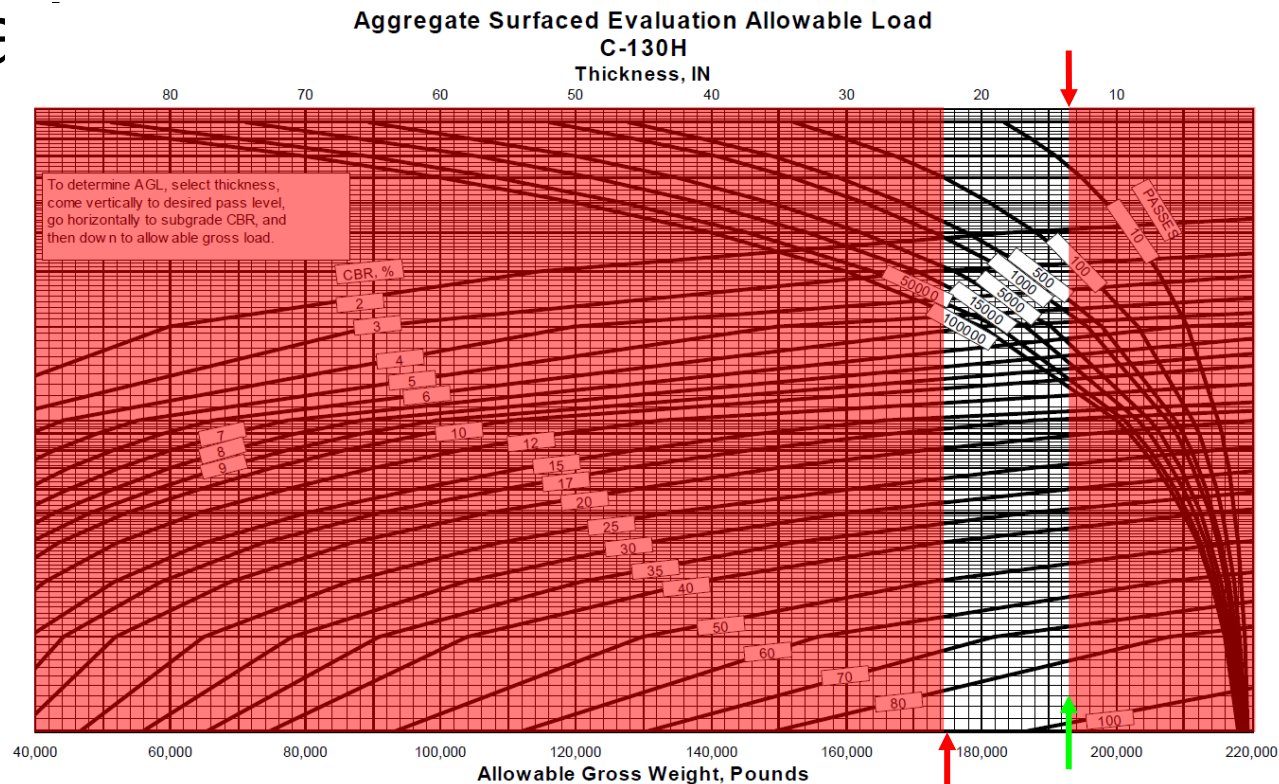


Base: 10k passes – Sub-base: 13.6", CBR 92 – Sub-grade: 13.8", CBR 39



Determining Passes

- What is the maximum allowable passes for a 175,000 lbs. KC-130 at the test loca

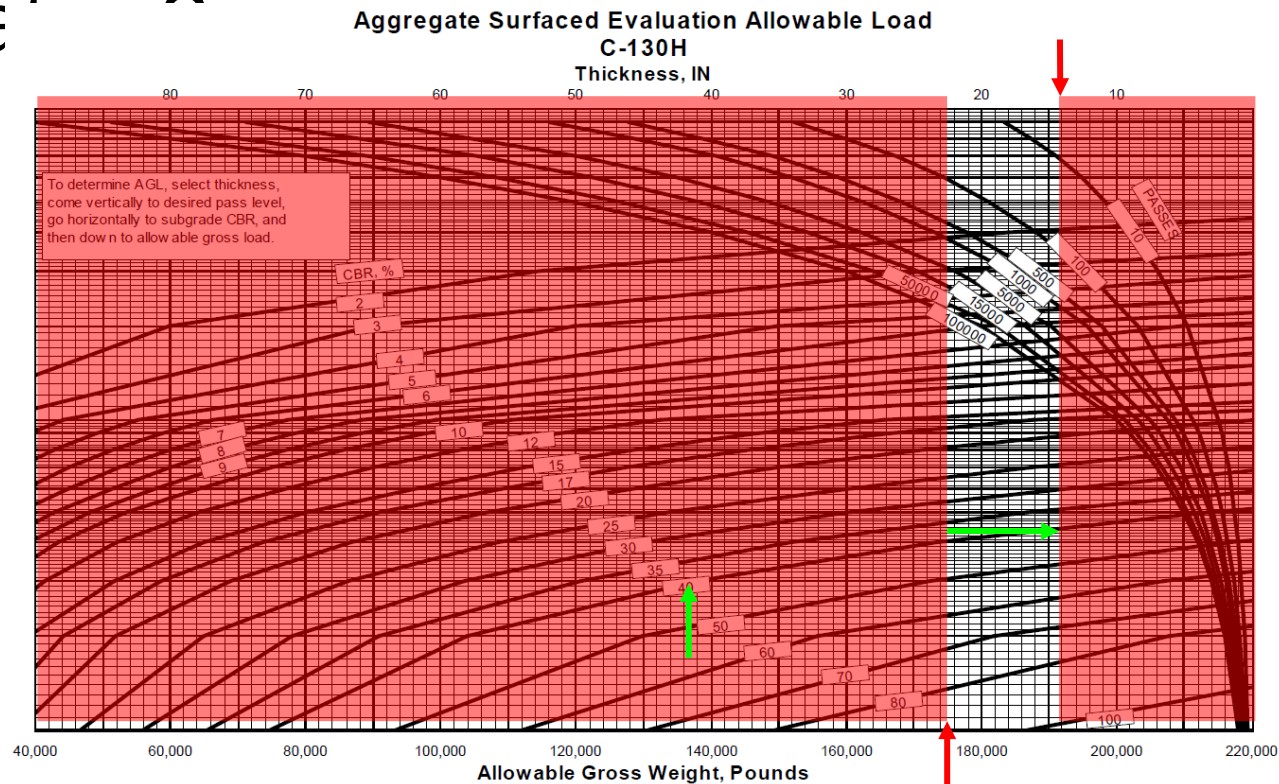


Base: 10k passes – Sub-base: 13.6" CBR 92 – Sub-grade: 13.8", CBR 39



Determining Passes

- What is the maximum allowable passes for a 175,000 lbs. KC-130 at the test location?



Base: 10k passes – Sub-base: 100k+ passes – Sub-grade: 13.8" CBR 39



Determining Passes

- What is the maximum allowable passes for a 175,000 lbs. KC-130 at the test location?
 - The weakest layer is the controlling layer
 - Maximum allowable passes is 10k

Base: 10k passes – Sub-base: 100k+ passes – Sub-grade: 100k+ passes



Comprehension Check

- How many test locations are used for a standard KC-130 semi-prepared runway?
- How do you determine the controlling point?
- How do you determine the controlling layer?
- How many tables do you need to determine the number of allowable passes?
- What information do you need prior to determining allowable passes?



Outline

- Identification
- Distresses
- Evaluation
- **Review**
- Summary



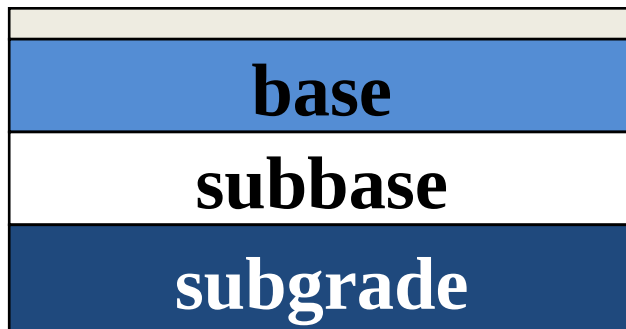
Review

- Identify soil classification, type, and characteristics
 - By weight: 12oz Gravels, 7oz Sands, 4oz Silts, 2oz Clays – Low Compressibility
 - ~ 50% Gravels, well-graded (GW)
 - By weight: 8oz Gravels, 12oz Sands, 2oz Silts, 1oz Clays – High Compressibility
 - ~ 50% Sands, poorly graded (SP)



Review

- Identify the different soil layers
- Assess CBR



No. of Blows	Accum Depth	Depth per Blow Set	Depth per Blow	CBR
0	2.5	0	0	N/A
4	3.7	1.2	.3	30
4	4.9	1.2	.3	30
4	6.1	1.2	.3	30
4	7.3	1.2	.3	30
5	8.4	1.1	.22	43
5	9.5	1.1	.22	43
7	10.6	1.1	.16	61
11	11.6	1.0	.1	100
11	12.6	1.0	.1	100
12	13.6	1.0	.1	100
12	14.6	1.0	.1	100
12	15.6	1.0	.1	100
11	16.6	1.0	.1	100
10	17.6	1.0	.1	100
10	18.6	1.0	.1	100
10	19.6	1.0	.1	100
6	20.7	1.1	.18	53
2	21.8	1.1	.55	15
2	23.0	1.2	.6	14
1	24.2	1.2	1.2	6
1	26.4	1.2	1.2	6
1	28.6	1.2	1.2	6
1	29.8	1.2	1.2	6
1	31.0	1.2	1.2	6
1	32.2	1.2	1.2	6
1	33.4	1.2	1.2	6
1	34.6	1.2	1.2	6
1	36.0	1.4	1.4	5

8.1" deep
33 blows
.24" per blow

10.1" deep
105 blows
.1" per blow

15.3" deep
13 blows
1.2" per blow

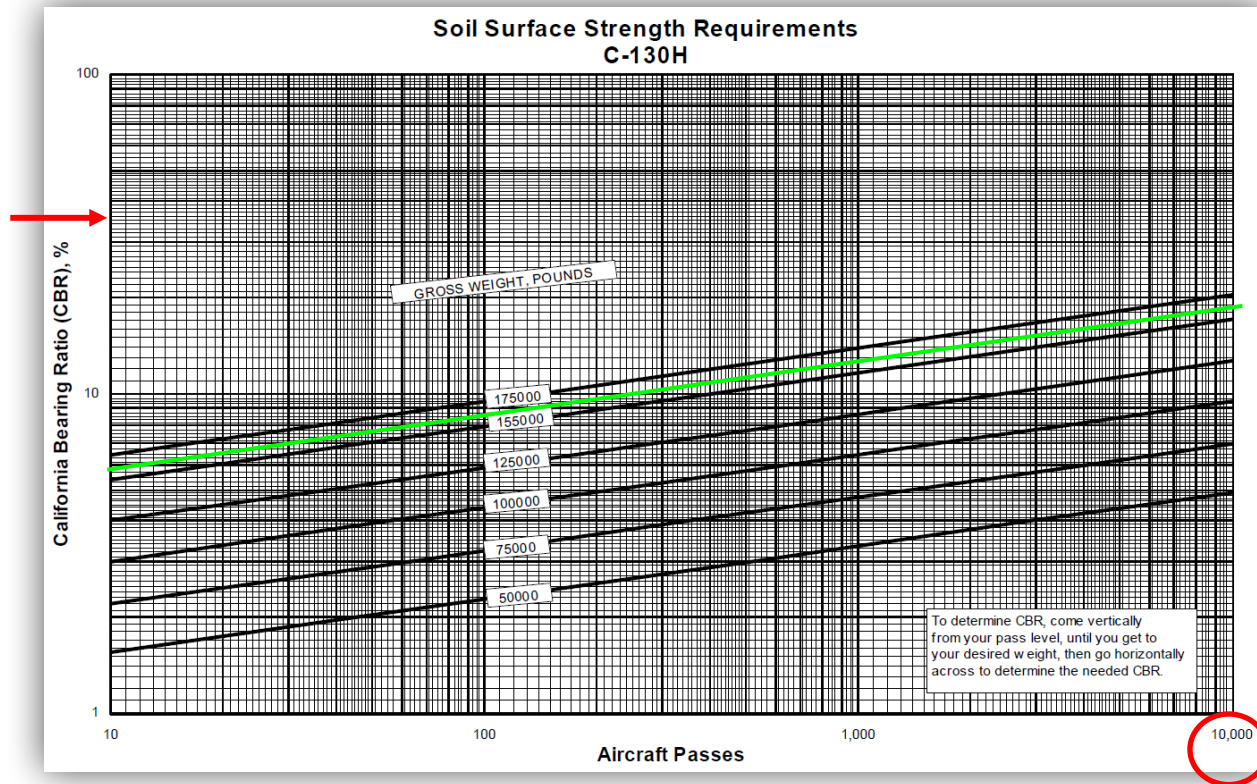
CBR Layer Structure

Layer	Thickness	CBR	Passes
Surface	2.5	N/A	N/A
Base	8.1	39	
Sub-Base	10.1	100	
Sub-grade	13.4	8	



Review

- Assess weight bearing capabilities for a 175,000 lbs. KC-130

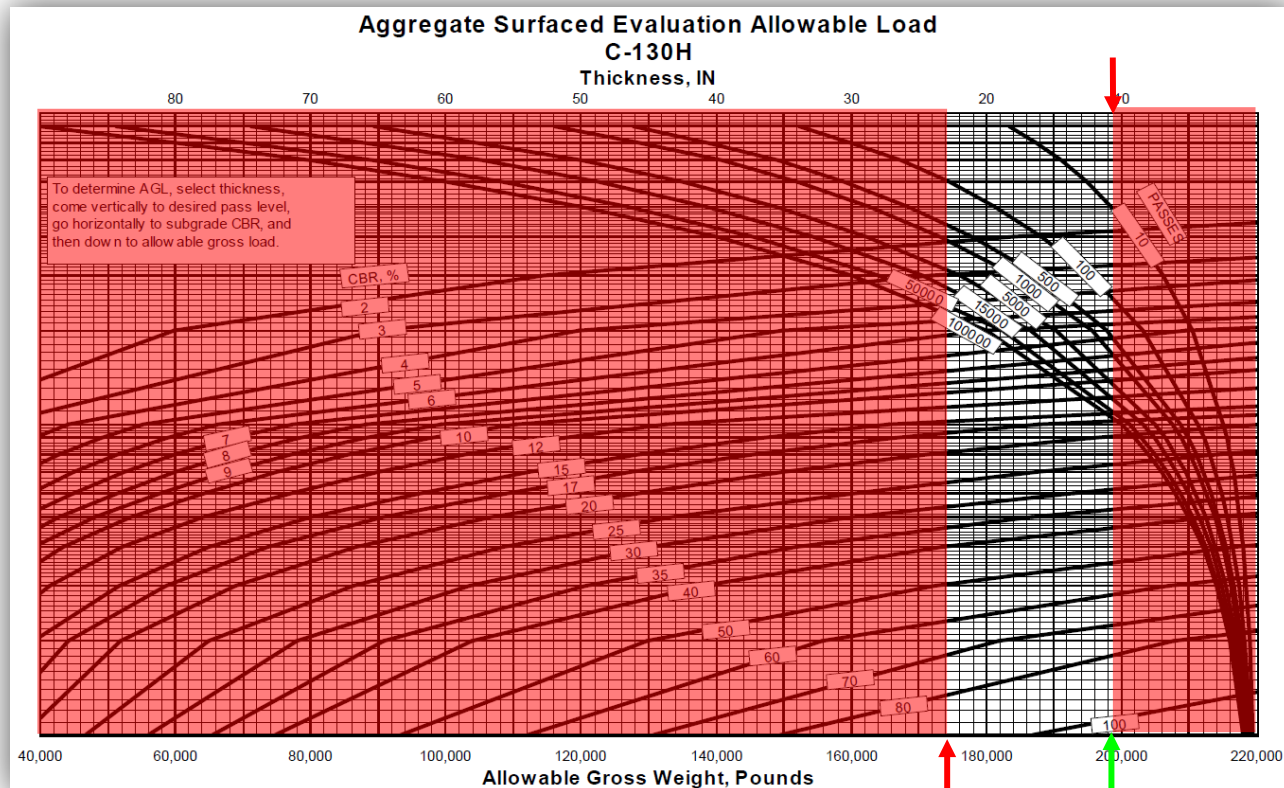


Base: 8.1", CBR 39 – Sub-base: 10.1", CBR 100 – Sub-grade: 15.3", CBR 6



Review

- Assess weight bearing capabilities for a 175,000 lbs. KC-130

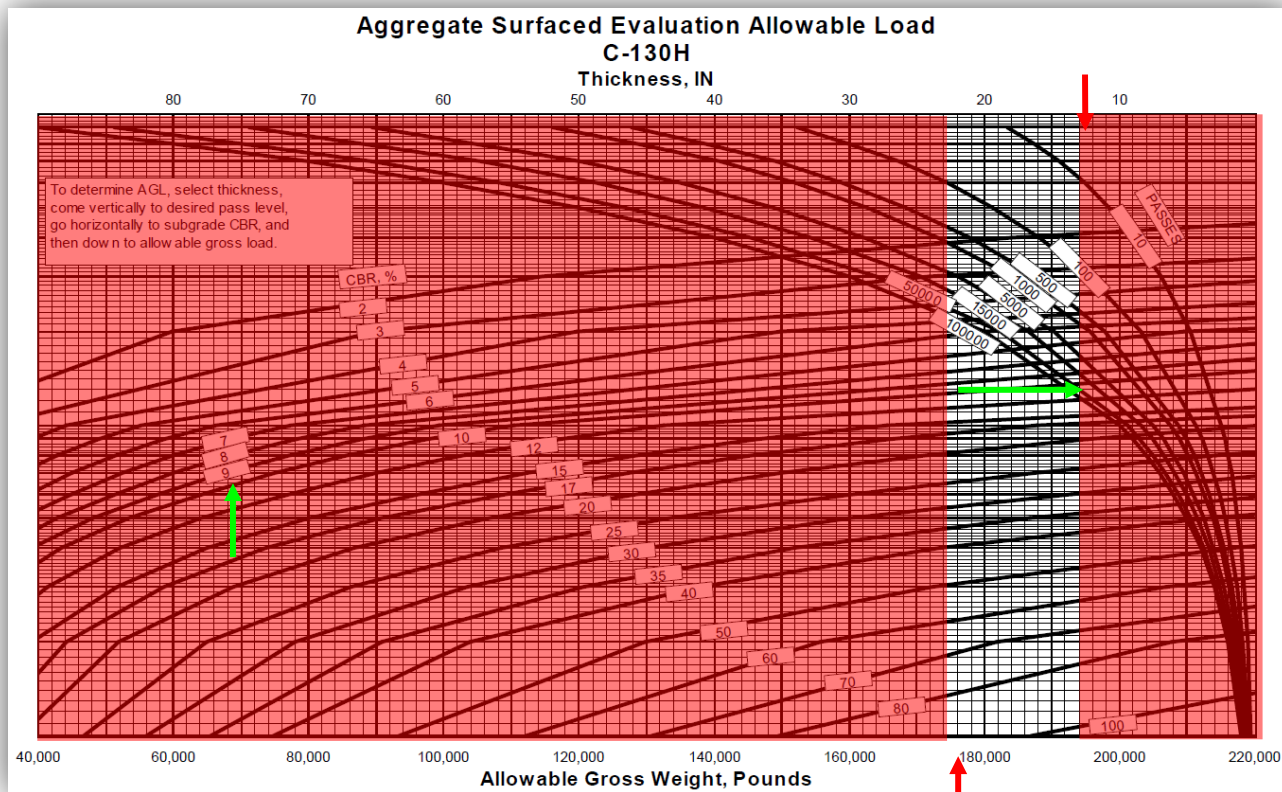


Base: 10k+ passes – Sub-base: 10.1", CBR 100 – Sub-grade: 15.3", CBR 6



Review

- Assess weight bearing capabilities for a 175,000 lbs. KC-130



Base: 10k+ passes – Sub-base: 100k+ passes – Sub-grade: 15.3" CBR 6



Review

- Assess weight bearing capabilities for a 175,000 lbs. KC-130
 - Base layer is the controlling layer
 - Maximum allowable passes is 10,000

Base: 10k+ passes – Sub-base: 100k+ passes – Sub-grade: ~15k passes



Outline

- Identification
- Distresses
- Evaluation
- Review
- Summary



Summary

- Identification
- Distresses
- Evaluation

Questions



Student Survey



This Presentation is Classified
UNCLASSIFIED